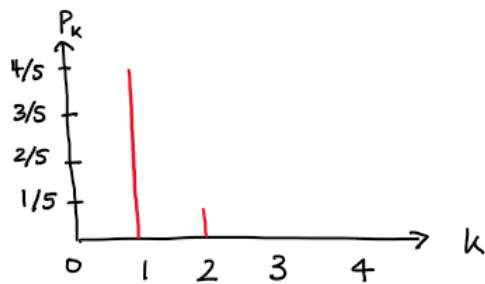


L3: Knowledge Check

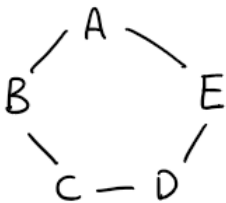
1

Multiple choice 1 point

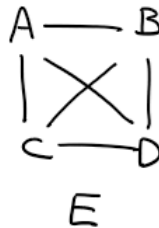


Which graph is consistent with the given empirical degree distribution?

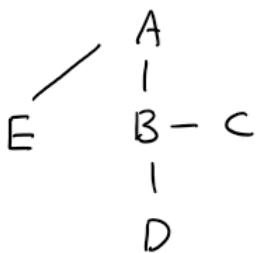
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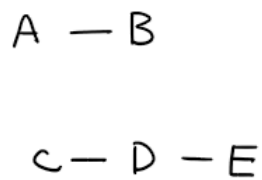
2.



3.



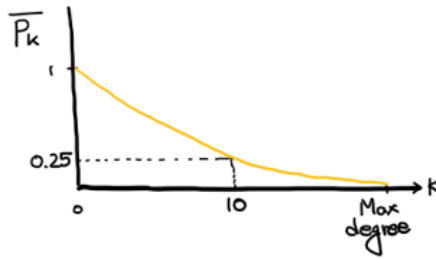
4.



- ☐ 4
- ☐ 3
- ☐ 2
- ☐ 1

2 Multiple choice 1 point

Which statement is true, based on the Complementary Cumulative Distribution Function (C-CDF) shown below?



- ☐ 75% of the nodes have degree less than 10
- ☐ The probability that a randomly chosen node has 10 neighbors is 25%
- ☐ $P(m)$ is 0 where m is the value of maximum degree in the graph
- ☐ None of the above

3 Multiple choice 1 point

Given a graph with 800 nodes and 1000 edges, what is the probability that a random edge connects to a node of degree 5? You are given that the probability that a node has degree 5 is $1/3$.

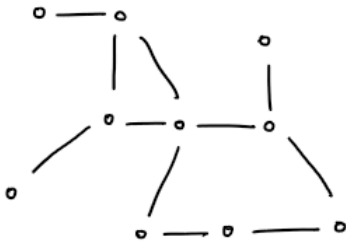
- ☐ $2/3$
- ☐ $1/3$
- ☐ $3/4$
- ☐ $4/5$

4

Multiple choice 1 point

In which of the following graphs Acquaintance Immunization would be **most** effective?

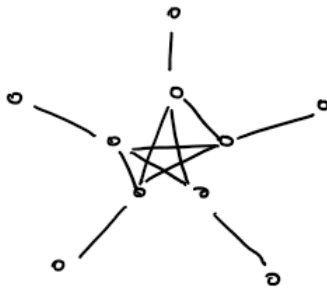
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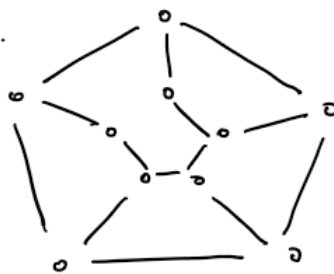
2.



3.



4.

☐ 2☐ 1☐ 4☐ 3

5

Multiple choice 1 point

Assume we can only vaccinate 100 people but we don't know the topology of the contact network. Which of the following strategies would be better to mitigate the spread of a virus?

- ☐ A. Select randomly 100 individuals and vaccinate them.
- ☐ B. Select randomly 100 individuals and ask each of them to report the neighbor with the highest number of contacts. Vaccinate the 100 reported nodes (i.e., you can assume that the reported nodes are distinct).
- ☐ C. Select randomly 100 individuals and ask each of them to report either the neighbor with the highest number of contacts – or themselves if they have more contacts than all their neighbors. Vaccinate the 100 reported nodes. (i.e., you can assume that the reported nodes are distinct).
- ☐ D. There is no difference between B and C.

6

Fill in the Blank 1 point

Consider the $G(n,p)$ model of random graphs with $n=1000$ nodes. What is the critical probability threshold for the emergence of a large connected component? Your answer should be in the form of a fraction (for example, $1/10$) -- do not make any approximations in this question.

7

Multiple choice 1 point

In the Friendship Paradox section of this lecture, we derived the average neighbor degree. What is the key assumption in those derivations?

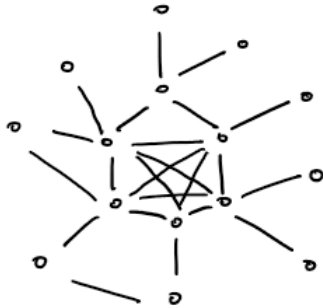
- ☐ The Friendship Paradox applies to any network
- ☐ The network is neutral
- ☐ The network follows the Poisson degree distribution
- ☐ The network is connected

8

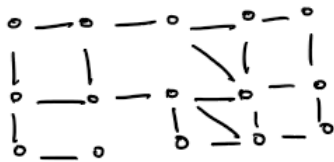
Multiple choice 1 point

Which of the following graphs would have the smallest correlation exponent value?

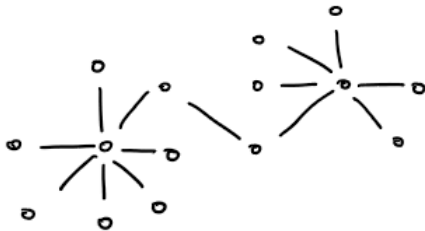
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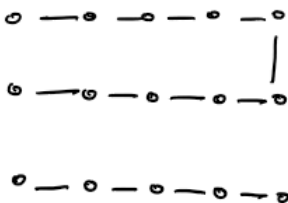
2.



3.



4.



- ☐ 3
- ☐ 2
- ☐ 1
- ☐ 4